



2014 TRIAL HIGHER SCHOOL CERTIFICATE

Mathematics

PM Friday 1 August

Section 1 – Multiple Choice

Sample $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9

A  B  C  D 

A B C D

A  B  C  D 

Start Here



- | | | | | |
|-----|-------------------------|-------------------------|-------------------------|-------------------------|
| 1. | A ☐ | B ☐ | C ☐ | D ☐ |
| 2. | A ☐ | B ☐ | C ☐ | D ☐ |
| 3. | A ☐ | B ☐ | C ☐ | D ☐ |
| 4. | A ☐ | B ☐ | C ☐ | D ☐ |
| 5. | A ☐ | B ☐ | C ☐ | D ☐ |
| 6. | A ☐ | B ☐ | C ☐ | D ☐ |
| 7. | A ☐ | B ☐ | C ☐ | D ☐ |
| 8. | A ☐ | B ☐ | C ☐ | D ☐ |
| 9. | A ☐ | B ☐ | C ☐ | D ☐ |
| 10. | A ☐ | B ☐ | C ☐ | D ☐ |

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Barker College

Student Number:

2014
TRIAL
HIGHER SCHOOL
CERTIFICATE

Mathematics

Staff Involved:

- JGD* • AJD
- LMD* • DZP
- BJB • GPF
- ASC • WMD
- PJR

PM Friday 1 August

135 copies

General Instructions

- Reading time - 5 minutes
- Working time - 3 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- Write your Barker Student Number on all pages of your solutions
- In Questions 11 – 16, show all relevant mathematical reasoning and/or calculations

Total marks – 100

Section I **Pages 2 - 4**

10 marks

- Attempt Questions 1 - 10
- Allow about 15 minutes for this section

Section II **Pages 5 - 12**

90 marks

- Attempt Questions 11 - 16
- Allow about 2 hours and 45 minutes for this section

Section 1 - Multiple Choice (10 marks)

Attempt Questions 1 - 10

Use the multiple-choice answer sheet for Questions 1 - 10

- 1 What is $\frac{2\sqrt{5}}{\sqrt{2} - \sqrt{5}}$ as a fraction with a rational denominator?

(A) $\frac{-5 - \sqrt{10}}{3}$ (B) $\frac{-5 + \sqrt{10}}{3}$ (C) $\frac{-10 + 2\sqrt{10}}{3}$ (D) $\frac{-10 - 2\sqrt{10}}{3}$

- 2 What is the value of $\sqrt{\frac{a^2 + b^2}{c^2}}$, if $a = 1.23$, $b = 0.85$ and $c = 4.81$?

Answer correct to three significant figures.

(A) 3.11 (B) 0.311 (C) 3.10 (D) 0.310

- 3 Given $\log_a 3 = 0.6$ and $\log_a 2 = 0.4$, find $\log_a 18$.

(A) 1.8 (B) 1.6 (C) 3.0 (D) 0.74

- 4 Evaluate $\sum_{r=3}^{10} 2r + 1$.

(A) 120 (B) 91 (C) 122 (D) 112

- 5 For $f(x) = \begin{cases} x+1 & x \geq 3 \\ x^2 + 2x - 1 & -1 < x < 3 \\ 3^x & x \leq -1 \end{cases}$, find the value of $f(2) + f(-2) - f(6)$.

(A) $\frac{1}{9}$ (B) $14\frac{1}{9}$ (C) 9 (D) $3\frac{8}{9}$

- 6 Evaluate $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1}$.
- (A) 2 (B) 1 (C) undefined (D) 0
- 7 What is the domain and range of the function $f(x) = |x - 2| + 1$?
- (A) Domain: $x \geq 0$, Range: $y \geq 0$
(B) Domain: $x < 2$, Range: $y \geq 1$
(C) Domain: All real x , Range: $y \geq 1$
(D) Domain: All real x , Range: $y \geq -1$
- 8 A primitive function of $4 + \sqrt{x}$ is
- (A) $4x + \frac{2\sqrt{x^3}}{3}$ (B) $4x + \frac{3\sqrt{x^2}}{2}$ (C) $4x + \frac{1}{2\sqrt{x}}$ (D) $\frac{1}{2\sqrt{x}}$
- 9 Given $\sin A = \frac{20}{29}$ and $\frac{\pi}{2} \leq A \leq \pi$, what is $\cos A$ in exact form?
- (A) $\frac{21}{29}$ (B) $\frac{-21}{29}$ (C) $\frac{21}{20}$ (D) $\frac{-21}{20}$
- 10 The derivative of $e^{\cos x}$ is
- (A) $\cos x \cdot e^{\sin x}$ (B) $-\cos x \cdot e^{\sin x}$ (C) $\sin x \cdot e^{\cos x}$ (D) $-\sin x \cdot e^{\cos x}$

End of Section I

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Section II – Extended Response (90 marks)

Attempt Questions 11 – 16

Allow about 2 hours and 45 minutes for this section

Answer each question on a separate writing booklet. Extra writing booklets are available.

In Questions 11 – 16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks)

[START A NEW BOOKLET]

Marks

(a) Simplify $\frac{y^3 - 8}{y - 2}$. 2

(b) Find the perpendicular distance from the line $y = 5x - 1$ to the point $(-3, 2)$. 2

(c) Solve $x^2 - 8x < 0$. 2

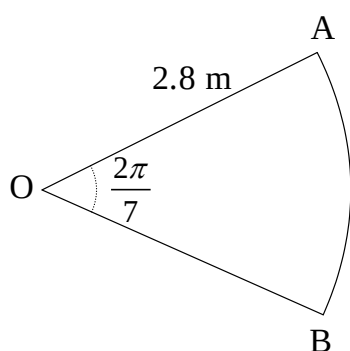
(d) Let α and β be the roots of the equation $x^2 - 2x - 1 = 0$. 2

What is the value of $\frac{1}{\alpha} + \frac{1}{\beta}$?

(e) In the diagram, AB is an arc of a circle with centre O. 2

The radius OA is 2.8 m and the angle AOB is $\frac{2\pi}{7}$ radians.

Find the area of the sector AOB.



(f) Find the exact value of $\cos \frac{\pi}{4} + \sin \frac{5\pi}{6}$. 2

(g) Solve $\ln 4 = 2 \ln x$. 3

End of Question 11

Question 12 (15 marks)

[START A NEW BOOKLET]

Marks(a) Differentiate with respect to x .

(i) $\sin(4x^2 + 1)$ 2

(ii) $x \ln x$ 2

(iii) $\frac{4x + 5}{2x^2 + 5x}$ 2

(b) Find

(i) $\int \sec^2 \frac{x}{2} dx$ 2

(ii) $\int_0^1 e^{2y+1} dy$ 2

(c) For the function $y = 3 - \cos 2x$

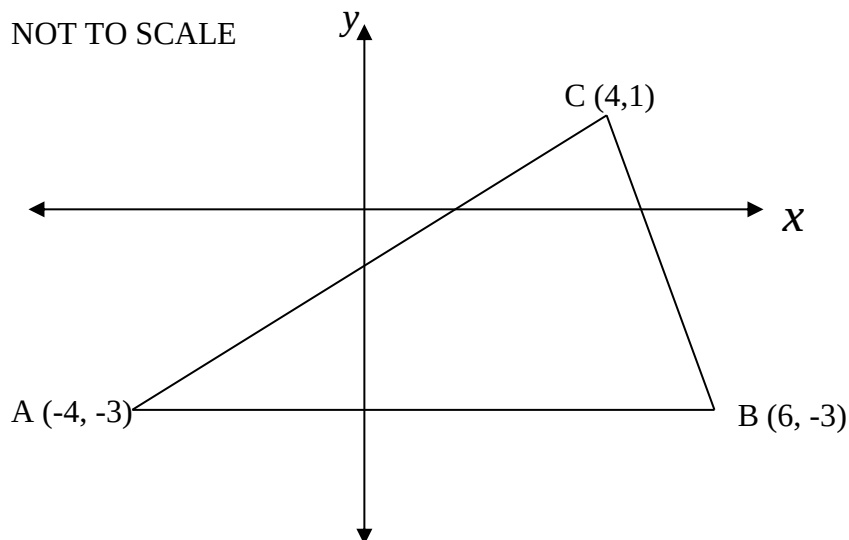
(i) State the period of the function. 1

(ii) State the amplitude of the function. 1

(iii) Sketch the graph of the function for $0 \leq \theta \leq 2\pi$. 3

End of Question 12

- (a) The diagram below shows $\triangle ABC$ where the vertices are $A(-4, -3)$, $B(6, -3)$ and $C(4, 1)$.



- | | | |
|-------|---|---|
| (i) | Find the gradient of the interval BC. | 1 |
| (ii) | Prove A, B and C are the vertices of a right-angled triangle. | 2 |
| (iii) | Find the distance BC in exact form. | 1 |
| (iv) | Find the size of $\angle CAB$ to the nearest minute. | 2 |
| (v) | Show that the coordinates of M, the midpoint of AB are $(1, -3)$. | 1 |
| (vi) | Show that the equation of the circle centre M, diameter AB is $x^2 + y^2 - 2x + 6y - 15 = 0$ | 3 |
| | | |
| (b) | (i) Sketch the curve $y = \ln(x+1)$ showing essential features. | 2 |
| | (ii) Using the Trapezoidal Rule with 4 sub-intervals, evaluate $\int_1^2 \ln(x+1) dx$ correct to 3 decimal places. | 2 |
| | (iii) Would the Trapezoidal Rule used in Part (ii) above provide a value that is greater or lesser than $\int_1^2 \ln(x+1) dx$? Explain your answer. | 1 |

End of Question 13

Question 14 (15 marks)

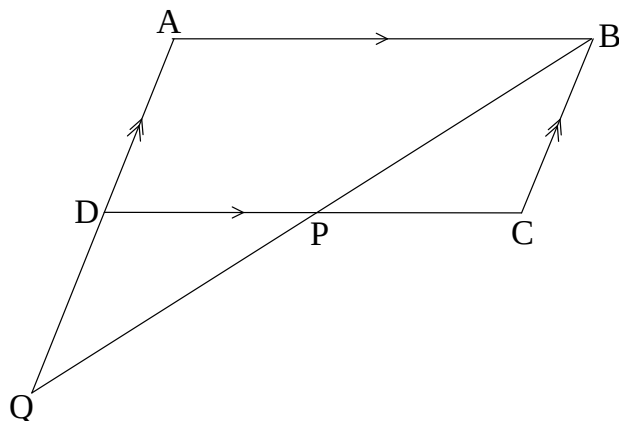
[START A NEW BOOKLET]

Marks

- (a) (i) For what values of x can the geometric series $1 + 2x + 4x^2 + 8x^3 + \dots$ have a limiting sum? **1**
- (ii) For what values of x does $\sum_{r=0}^{\infty} (2x)^r = \frac{5}{9}$? **2**
- (b) Simplify $\frac{2\sec^2 A - 2}{4 \tan A}$. **2**
- (c) What are the coordinates of the focus of the parabola defined by the equation $y = -16x^2$? **2**
- (d) For the curve $f(x) = \frac{2}{3}x^3 - 8x + 1$,
- (i) Find the coordinates of the stationary point(s) and determine their nature. **3**
- (ii) Determine any point(s) of inflection. **2**
- (iii) Sketch the curve, showing all of the above features. **2**
- (iv) State the domain of x for which $f(x)$ is monotonic increasing. **1**

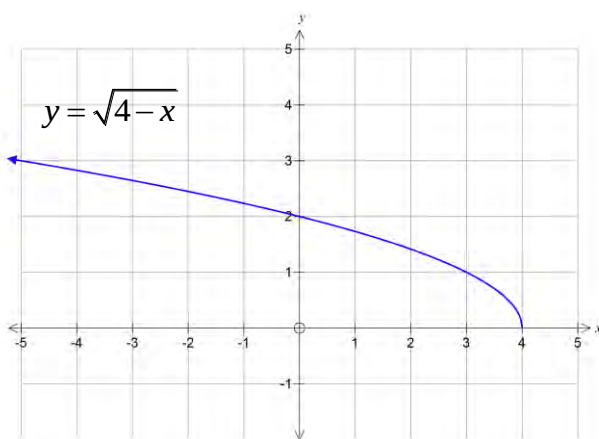
End of Question 14

- (a) ABCD is a parallelogram. $DP = PC$ and P lies on DC. BP produced, meets AD produced at Q.



- (i) Prove that $\triangle DPQ \equiv \triangle CPB$. 2
- (ii) Hence, or otherwise, show that $BP = PQ$. 1
- (b) (i) Find the points of intersection of the curves $y = \frac{1}{2-x}$ and $y = 2-x$. 3
- (ii) On the same large diagram, sketch and label these two curves showing any asymptotes and intercepts with the coordinate axes. 3
- (iii) Find the area bounded by the two curves and the y-axis. 3

- (c) Find the volume of the solid formed when the area bounded by $y = \sqrt{4-x}$, $y = 0$ and the y -axis is rotated around the x -axis. 3



End of Question 15

- (a) A Year 12 Biology student tested to see how much bacteria was present in a variety of food samples left in the classroom.

It is known that after t hours the number of bacteria (N) present in a particular type of food is given by the formula $N = N_0 e^{kt}$.

- (i) If initially there were 20 000 bacteria present and after three hours there were 45 000 bacteria present, calculate the value of k (correct to 2 decimal places). 1

- (ii) How long would it take for the initial number of bacteria to triple in quantity? 2

- (iii) What will be the rate of increase of the bacteria after $4\frac{1}{2}$ hours? 2

- (b) Jessica plans to work for the next ten years.

During that time she wants to save \$50 000.

She decides to invest a fixed amount of money at the beginning of each month during this time.

Interest is paid at a fixed rate of 6% p.a. **compounded monthly**.

- (i) Let M be the monthly investment in dollars. 2

Show that the total investment R after 10 years is given by

$$R = M \left[(1.005) + (1.005)^2 + (1.005)^3 + \dots + (1.005)^{120} \right].$$

- (ii) Find the amount M needing to be deposited each month to reach her goal. 3

Question 16 (continued)**Marks**

- (c) A 6 metre piece of wire is cut into three pieces to form two congruent squares and a circle of radius r metres.

- (i) Show that the total combined area of these three shapes is given by

2

$$A = \frac{(3 - \pi r)^2}{8} + \pi r^2.$$

- (ii) Find the exact value of r which will make this total combined area a minimum.

3**End of Paper**

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \quad \text{if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax \, dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax \, dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax \, dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax \, dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

HSC
2014 Year 12 Mathematics
Trial Examination.

$$1. \frac{2\sqrt{5}}{\sqrt{2}-\sqrt{5}} \times \frac{\sqrt{2}+\sqrt{5}}{\sqrt{2}+\sqrt{5}} = \frac{2\sqrt{10}+10}{2-5} = \frac{2\sqrt{10}+10}{-3}$$

$$= \frac{-2\sqrt{10}-10}{3} \quad (D)$$

$$2. \sqrt{\frac{1 \cdot 23^2 + 0 \cdot 85^2}{4 \cdot 81^2}} = 0.3108368842$$

$$= 0.311 \quad (3 \text{ sig figs}) \quad (B)$$

$$3. \log_a 18 = \log_a (3^2 \times 2)$$

$$= \log_a 3^2 + \log_a 2$$

$$= 2\log_a 3 + \log_a 2$$

$$= 2 \times 0.6 + 0.4 = 1.6 \quad (B)$$

$$4. \sum_{r=3}^{10} 2r+1 = 7+9+11+\dots+21$$

$$= 112 \quad (D)$$

$$5. f(2) = 2^2 + 2 \times 2 - 1 = 7$$

$$f(-2) = 3^{-2} = \frac{1}{9}$$

$$f(6) = 7$$

$$\therefore f(2) + f(-2) - f(6) = 7 + \frac{1}{9} - 7 = \frac{1}{9} \quad (A)$$

$$6. \lim_{x \rightarrow 1} \frac{x^2-1}{x-1} = \lim_{x \rightarrow 1} \frac{(x-1)(x+1)}{x-1}$$

$$= \lim_{x \rightarrow 1} (x+1)$$

$$= 1+1 = 2 \quad (A)$$

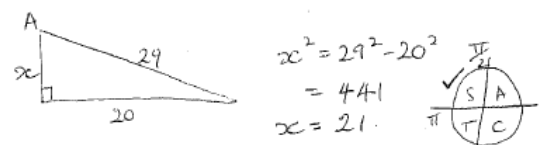
7. (C)

$$8. y' = 4 + \sqrt{x}$$

$$= 4 + x^{\frac{1}{2}}$$

$$y = 4x + \frac{2x^{\frac{3}{2}}}{\frac{3}{2}}$$

$$= 4x + \frac{2\sqrt{x^3}}{3} \quad (A)$$

9. 

$$x^2 = 29^2 - 20^2$$

$$= 441$$

$$x = 21$$

$$\therefore \cos A = -\frac{21}{29} \quad (B)$$

10. $y = e^{\cos x}$

$$y' = -\sin x \cdot e^{\cos x} \quad (D)$$

11. a) $\frac{y^3-8}{y-2} = \frac{(y-2)(y^2+2y+4)}{y-2}$

$$= y^2+2y+4$$

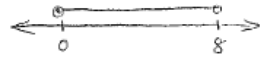
b) $5x - y - 1 = 0 \quad (-3, 2)$

$$a=5 \quad b=-1 \quad c=-1 \quad x_1 = -3 \quad y_1 = 2$$

$$Ld = \frac{|5x-3+(-1)x-2-1|}{\sqrt{5^2+1^2}}$$

$$= \frac{|1-15-2-1|}{\sqrt{26}} = \frac{18}{\sqrt{26}} \text{ units.}$$

11 c) $x^2 - 8x < 0$
 $x(x-8) < 0$



$$0 < x < 8$$

d) $x^2 - 2x - 1 = 0$

$$\alpha + \beta = 2 \quad \alpha\beta = -1$$

$$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\beta + \alpha}{\alpha\beta} = \frac{2}{-1} = -2$$

e) $A = \frac{1}{2} r^2 \theta$

$$= \frac{1}{2} \times 2 \cdot 8^2 \times \frac{2\pi}{7}$$

$$= 3.52 \text{ m}^2 \text{ OR } 1.12\pi \text{ m}^2$$

f) $\cos \frac{\pi}{4} + \sin \frac{5\pi}{6} = \frac{1}{\sqrt{2}} + \sin \frac{\pi}{6}$



$$= \frac{1}{\sqrt{2}} + \frac{1}{2}$$

$$= \frac{\sqrt{2}}{2} + \frac{1}{2}$$

$$= \frac{1+\sqrt{2}}{2}$$

g) $\ln 4 = 2 \ln x$

$$\ln 4 = \ln x^2$$

$$4 = x^2$$

$$x = \pm 2$$

But $x > 0 \therefore x = 2$
 is the only solution.

12. a) (i) $y = \sin(4x^2 + 1)$
 $y' = 8x \cos(4x^2 + 1)$

(ii) $y = x \ln x$
 $y' = \ln x + x \times \frac{1}{x}$
 $= \ln x + 1$

(iii) $y = \frac{4x+5}{2x^2+5x}$

$$y' = \frac{(2x^2+5) \times 4 - (4x+5)(4x+5)}{(2x^2+5x)^2}$$

$$= \frac{4(2x^2+5) - (4x+5)^2}{(2x^2+5x)^2}$$

b) (i) $\int \sec^2 \frac{x}{2} dx$

$$= 2 \tan \frac{x}{2} + C$$

(ii) $\int_0^1 e^{2y+1} dy$

$$= \left[\frac{e^{2y+1}}{2} \right]_0^1$$

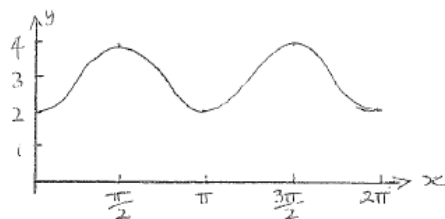
$$= \frac{e^3}{2} - \frac{e}{2}$$

$$= \frac{e^3 - e}{2}$$

c) (i) Period = $\frac{2\pi}{2} = \pi$

(ii) Amplitude = 1

12 c) (iii)



$$13 a) (i) m_{BC} = \frac{-3-1}{6-4} = \frac{-4}{2} = -2$$

$$(ii) m_{AC} = \frac{-3-1}{-4-4} = \frac{-4}{-8} = \frac{1}{2}$$

$$m_{BC} \times m_{AC} = -2 \times \frac{1}{2} = -1$$

$\therefore AC \perp BC$

$\therefore \triangle ABC$ is right-angled

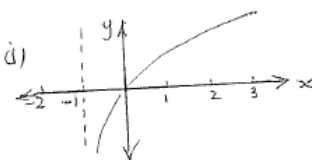
$$(iii) \text{dist}_{BC} = \sqrt{(6-4)^2 + (-3-1)^2} \\ = \sqrt{2^2 + 4^2} \\ = \sqrt{20} = 2\sqrt{5}$$

$$(iv) \sin \theta = \frac{2\sqrt{5}}{10} = \frac{\sqrt{5}}{5} \\ \theta = 26^\circ 33' 54.18'' \\ = 26^\circ 34'$$

$$(v) m_{P_{AB}} = \left(\frac{-4+6}{2}, \frac{-3-3}{2} \right) \\ = \left(\frac{2}{2}, -\frac{6}{2} \right) = (1, -3)$$

$$(vi) (x-1)^2 + (y+3)^2 = 25 \\ x^2 - 2x + 1 + y^2 + 6y + 9 = 25 \\ x^2 + y^2 - 2x + 6y - 15 = 0$$

13 b) (i)



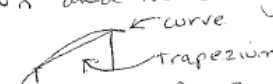
$$(ii) \begin{array}{c|c|c|c|c|c} x & 1 & \frac{5}{4} & \frac{3}{2} & \frac{7}{4} & 2 \\ \hline y & \ln 2 & \ln \frac{9}{4} & \ln \frac{5}{2} & \ln \frac{11}{4} & \ln 3 \end{array}$$

$$\int \ln(x+1) dx \div \frac{0.25}{2} (\ln 2 + 2 \times \ln \frac{9}{4} + 2 \times \ln \frac{5}{2} + 2 \times \ln \frac{11}{4} + \ln 3)$$

$$= 0.9086753986$$

$$= 0.909 \text{ units}^2$$

(iii) Less because the curve is concave down and the straight line cuts below



$$14 a) (i) 1 + 2x + 4x^2 + 8x^3 + \dots$$

$$a=1 \quad r=2x$$

For a limiting sum $-1 < r < 1 \quad r \neq 0$

$$\therefore -1 < 2x < 1$$

$$-\frac{1}{2} < x < \frac{1}{2} \quad x \neq 0$$

$$(ii) S_{\infty} = \frac{a}{1-r}$$

$$\frac{5}{9} = \frac{1}{1-2x}$$

$$5(1-2x) = 9$$

$$5 - 10x = 9$$

$$-10x = 4$$

$$x = -\frac{2}{5}$$

$$14 \text{ b) } \frac{2\sec^2 A - 2}{4 + \tan A} = \frac{2(\sec^2 A - 1)}{4 + \tan A}$$

$$= \frac{2 \tan^2 A}{4 + \tan A}$$

$$= \frac{\tan A}{2}$$

$$c) x^2 = \frac{1}{16} y$$

$$4a = \frac{1}{16}$$

$$a = \frac{1}{64}$$

Focus is at $(0, -\frac{1}{64})$

$$d) f(x) = \frac{2}{3} x^3 - 8x + 1$$

$$i) f'(x) = 2x^2 - 8$$

stationary points when $f'(x) = 0$

$$0 = 2x^2 - 8$$

$$= x^2 - 4$$

$$= (x-2)(x+2)$$

$$\therefore x = \pm 2$$

$$\begin{array}{c|ccc|ccc} x & -3 & -2 & 0 & 2 & 3 \\ f'(x) & 10 & 0 & -8 & 0 & 10 \\ & / & - & \backslash & - & / \end{array}$$

$$\text{OR } f''(x) = 4x$$

when $x = -2$ $f''(x) = -8$ $\therefore$ concave down $\cap$ max T.P.

when $x = 2$ $f''(x) = 8$ $\therefore$ concave up $\cup$ min T.P.

$$\begin{array}{l} (-2, \frac{35}{3}) \text{ max T.P.} \\ (2, -\frac{29}{3}) \text{ min T.P.} \end{array}$$

14 d) (ii) Possible POI when $f''(x) = 0$

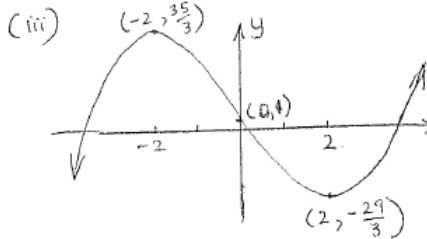
$$f''(x) = 4x$$

$$4x = 0$$

$$x = 0$$

$$\begin{array}{c|ccc} x & -1 & 0 & 1 \\ f''(x) & -4 & 0 & 4 \\ & \cap & & \cup \end{array}$$

$\therefore (0, 1)$ is a point of inflexion.



$$(iv) x < -2 \text{ or } x > 2$$

15 a) i) In $\triangle DPQ$ and $\triangle CPB$

$\angle QDP = \angle BCP$ (alternate angles on parallel lines $AQ \parallel BC$)

$DP = PC$ (given)

$\angle DPQ = \angle CPB$ (vertically opposite angles)

$\therefore \triangle DPQ \cong \triangle CPB$ (AAS)

(ii) $BP = PQ$ (corresponding sides in congruent $\triangle$ s)

$$b) y = \frac{1}{2-x} \quad \left\{ \quad \frac{1}{2-x} = 2-x \right.$$

$$y = 2-x \quad \left\{ \quad 1 = (2-x)(2-x) \right.$$

$$1 = 4 - 4x + x^2$$

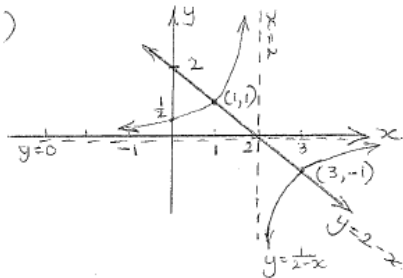
$$x^2 - 4x + 4 - 1 = 0$$

$$x^2 - 4x + 3 = 0$$

$$(x-3)(x-1) = 0$$

$$\therefore (3, -1) \quad (-1, 1)$$

15 b) (i)



$$\begin{aligned}
 \text{(iii)} \quad A &= \frac{1}{2}(2+1) \times 1 - \int_0^1 \frac{1}{2-x} dx \\
 &= \frac{3}{2} - 1 [\ln(2-x)]_0^1 \\
 &= \frac{3}{2} + [\ln 1 - \ln 2] = \frac{3}{2} + \ln \frac{1}{2} \\
 &= \frac{3}{2} - 0.6931471806 = \frac{3}{2} - \ln 2 \\
 &= 0.8068528194 \text{ units}^2
 \end{aligned}$$

$$c) V = \int_0^4 \pi y^2 dx$$

$$V = \pi \int_0^4 4-x dx$$

$$= \pi \left[4x - \frac{x^2}{2} \right]_0^4$$

$$= \pi [(16-8) - (0-0)]$$

$$= 8\pi \text{ units}^3 = 25.13274123 \text{ units}^3$$

$$16. a) i) N = N_0 e^{kt}$$

$$\text{when } t=0, N=20000 \therefore N_0 = 20000$$

$$\text{when } t=3, N=45000$$

$$45000 = 20000 e^{3k}$$

$$\frac{45}{20} = e^{3k}$$

$$\ln\left(\frac{9}{4}\right) = 3k \quad k = \frac{\ln \frac{9}{4}}{3} = 0.2703100721 \approx 0.27$$

$$ii) 60000 = 20000 e^{0.27t}$$

$$3 = e^{0.27t}$$

$$\ln 3 = 0.27t$$

$$t = \frac{\ln 3}{0.27} = 4.068934402 \text{ hours}$$

$$= 4 \text{ hours } 4 \text{ mins } 8.16 \text{ seconds}$$

$$iii) \text{ When } t=4.5 \quad \frac{dN}{dt} = 0.27 \times 20000 e^{0.27t}$$

$$\frac{dN}{dt} = 0.27 \times 20000 \times e^{0.27 \times 4.5}$$

$$= 18199.58795$$

$$= 18200 \text{ bacteria/hr at the 4.5 hour mark.}$$

$$b) 50000 \quad 6\% \text{ pa} = 0.5\% \text{ per month}$$

$$i) A_1 = M \times 1.005$$

$$A_2 = (A_1 + M) \times 1.005$$

$$= (1.005M + M) \times 1.005$$

$$= 1.005^2 M + 1.005M$$

$$A_3 = (A_2 + M) \times 1.005$$

$$= ((1.005^2 M + 1.005M) + M) \times 1.005$$

$$= 1.005^3 M + 1.005^2 M + 1.005M$$

$$A_{120} = R = 1.005^{120} M + 1.005^{119} M + \dots + 1.005M$$

$$= M(1.005 + 1.005^2 + \dots + 1.005^{119} + 1.005^{120})$$

16 b) (ii) $1.005 + 1.005^2 + \dots + 1.005^{120}$
 $a = 1.005$ $r = 1.005$ $n = 120$
 $R = m \times \left(\frac{1.005(1.005^{120} - 1)}{1.005 - 1} \right)$
 $50000 \times (1.005 - 1) = m(1.005(1.005^{120} - 1))$
 $m = \frac{50000 \times (1.005 - 1)}{1.005(1.005^{120} - 1)}$
 $= 303.5845868 = \$303.58$

c)  $C = 2\pi r$ $A = \pi r^2$
 $\frac{6 - 2\pi r}{2} = \text{one square perimeter}$

$P = 3 - \pi r$

side of each square $= \frac{3 - \pi r}{4}$

area of each square
 $= \left(\frac{3 - \pi r}{4} \right)^2$

$\therefore$ Total area $= 2 \times \left(\frac{3 - \pi r}{4} \right)^2 + \pi r^2$
 $= \frac{2 \times (3 - \pi r)^2}{16} + \pi r^2$
 $= \frac{(3 - \pi r)^2}{8} + \pi r^2$

Minimum when $A' = 0$.

$A' = \frac{2(3 - \pi r)}{8} \times -\pi + 2\pi r$

$0 = -\frac{\pi(3 - \pi r)}{4} + 2\pi r$

$0 = -3\pi + \pi^2 r + 8\pi r$
 $= -3 + \pi r + 8r$

$3 = (\pi + 8)r$ $r = \frac{3}{\pi + 8} \approx 0.269$

x	0	$\frac{3}{\pi+8}$	1
A'	$-\frac{3\pi}{4}$	0	$-\frac{\pi(3-\pi)}{4} + 2\pi \approx 6.31$

$\therefore$ minimum when $r = \frac{3}{\pi+8}$